University of Barishal

Department of Computer Science and Engineering

Final Exam - Semester: 6th

Course: Course Title: Peripherals and Interfacing (CSE-3207) **Session:** 20-21

Time: 180 minutes | **Marks:** 60.00 **Teacher:** Dr. Tania Islam

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> *Answer any five(5) from the followings*

1.

a) What is a peripheral device in a computer system? Why do we need interfacing circuits between the CPU and peripherals?

b) Explain why peripherals cannot be directly connected to a microprocessor. Describe how an interface unit helps in matching speed, data format, and voltage levels between the CPU and external devices. Give one simple example.

c) Differentiate between off-line and on-line peripherals.

2.

a) Explain what Bluetooth technology is, how it connects devices wirelessly, and give examples (headphones, keyboards, file transfer between mobiles).

b) Describe why USB is widely used, its advantages (plug-and-play, high speed, power supply through cable), and examples of USB devices.

c) Explain in simple terms how data flows through USB (host controller → device → packets), how the device is detected, and how communication is established

3.

a) Explain from the figure, how the control unit sends signals to the datapath, memory, and I/O devices. Mention how it controls reading, writing, and executing instructions based on the system bus.

b) Explain what the datapath contains (ALU, registers, buses) and what the control unit does (generates control signals, manages instruction execution). Describe how both work together.

c) Describe why we separate hardware into layers (CPU, memory, I/O), how abstraction hides low-level details, and how it helps software work on different hardware without changes.

4.

a) What is the role of the Accumulator (ACC) and ALU as shown in the 8085 architecture diagram?

b) Why is the Instruction Register (IR) and Instruction Decoder important in the 8085 architecture as shown in the figure?

c) How does the 8085 use internal buses and registers (W, Z, B, C, D, E, H, L, SP, PC) shown in the figure for data transfer?

5.

a) Why must the order of PUSH and POP operations be opposite when using the stack in a microprocessor?

b) How does the 8085 microprocessor use the stack and stack pointer during CALL and RET instructions?

c) What is the difference between call by value and call by reference, and how is the stack used in these methods?

6.

a) Explain the concept of superscalar processing. Why is the 8085 microprocessor not considered a superscalar processor? Discuss any two features that differentiate a superscalar CPU from the 8085.

b) Describe the superscalar instruction decode and issue mechanism. How does dependency analysis (RAW, WAR, WAW) help the processor decide whether instructions can be issued in parallel?

c) What is parallel encoding in instruction processing? Explain how parallel encoding and decoding enable multiple instruction pipelines. Give an example of independent and dependent instruction pairs.

7.

a) Define a RISC processor. List any four characteristics that differentiate it from a CISC processor.

b) Explain the purpose of the Program Counter (PC) in assembly language programming. How does it work in a RISC processor?

c) What is an opcode in assembly language? Give an example of a RISC instruction and identify its opcode.

8.

a) What is a peripheral device? Give two examples of input devices and two examples of output devices.

b) Explain the difference between memory-mapped I/O and port-mapped I/O.

c) What is an interrupt? Explain how it helps in interfacing peripherals with a microprocessor.